Petroleum Engineering 324
Well Performance
Pseudosteady-State Flow in a Circular Reservoir

Growth is the only evidence of life.
— John Henry Newman (1864)

Topic: Pseudosteady-State Flow in a Circular Reservoir

Objectives: (things you should know and/or be able to do)

- Be familiar with and be able to derive the single-phase, pseudosteady-state flow relations for compressible liquids in a radial flow system. In particular, you should be able to derive the following:
  - p_r-p_{wf} flow relation:

$$p_r = p_{wf} + \frac{1}{c_r}\frac{qB\mu}{kh}\left[\frac{r_e^2}{(r_e^2-r_w^2)}\ln\left[\frac{r}{r_w}\right] - \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)} + s\right]$$

  - $\bar{p}$-p_{wf} flow relation:

$$\bar{p} = p_{wf} + \frac{1}{c_r}\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4} + s\right] \qquad \text{(Well Centered in a Circular Reservoir)}$$

$$\bar{p} = p_{wf} + \frac{1}{c_r}\frac{qB\mu}{kh}\left[\frac{1}{2}\ln\left[\frac{4}{e^{\gamma}}\frac{A}{r_w^2}\frac{1}{C_A}\right] + s\right] \qquad \text{(General Formulation)}$$

  where:

  γ = 0.577216... (Euler's Constant)
  C_A = Dietz "shape factor" (*e.g.*, C_A = 31.62 for a well in a circular reservoir)

  - $p(r,t)$ solution for pseudosteady-state flow conditions:

$$p_r = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)} - \frac{3}{4}\right] - \frac{qB}{V_p c_t}t \qquad \text{(Darcy Units)}$$

$$p_r = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)} - \frac{3}{4}\right] - 5.615\frac{qB}{V_p c_t}t \qquad \text{(Field Units)}$$

  where for field units we use t in days, and V_p in ft^3.

  - $p(r_w,t)$ solution for pseudosteady-state flow conditions:

$$p_{wf} = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4} + s\right] - 5.615\frac{qB}{V_p c_t}t \qquad \text{(Field Units)}$$

Table of Units Conversions: (for the equations given above)

Variable	Darcy Units	Field Units	SI Units
c_r	2π	$2\pi 1.127\text{x}10^{-3}$ or $7.081\text{x}10^{-3}$	$2\pi 8.527\text{x}10^{-5}$ or $5.358\text{x}10^{-4}$

- Be able to derive the $r(\bar{p})$ result for a radial system and be able to relate this result to reservoir performance.
- Be able to sketch pressure distributions during steady-state and pseudosteady-state flow conditions for a radial system.

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Lecture Outline:

- Development of pseudosteady-state flow relations (attached notes)
  - Material balance considerations.
  - p_r-p_{wf} flow relation.
  - $\bar{p}$-p_{wf} flow relation.
  - $p(r,t)$ solution for pseudosteady-state flow conditions.
- Review illustrative plots of pseudosteady-state flow performance.

Reading Assignment:

1. Dake, L.P.: "*Fundamentals of Reservoir Engineering*," Elsevier Scientific Publishing Company, 1978.
   - Chapter 6—Well Inflow Equations for Stabilized Flow Conditions.

- Review the attached notes.
  - Derivation of the Pseudosteady-State Flow Relations for a Radial System.
  - Illustrations of Pseudosteady-State Performance in Radial Flow Systems.
  - Derivation of the Pseudosteady-State Flow Relations for a Radial System.
- Dietz, D.N.: "Determination of Average Reservoir Pressure From Buildup Surveys," *JPT* (Aug. 1965) 955-959.
- Blasingame, T.A. and Lee, W.J.: "Properties of Homogeneous Reservoirs, Naturally Fractured Reservoirs, and Hydraulically Fractured Reservoirs from Decline Curve Analysis," paper SPE 15018 presented at the SPE Permian Basin Oil and Gas Recovery Conference, Midland, TX, 13-14 March 1986.
- Blasingame, T.A. and Lee, W.J.: " Variable-Rate Reservoir Limits Testing," paper SPE 15028 presented at the SPE Permian Basin Oil and Gas Recovery Conference, Midland, TX, 13-14 March 1986.

Derivation of the Pseudosteady-State Flow Relations for a Radial System

- Physical Considerations
- Material Balance Considerations
- Pseudosteady-State Solutions of the Radial Flow Diffusivity Equation
  - p_r-p_{wf} Formulation
  - $\bar{p}$-p_{wf} Formulation
  - $\bar{p}$-$\bar{r}$ Concept (*i.e.*, $r(\bar{p})$)
  - $p(r,t)$ Solution for Pseudosteady-State Flow Conditions

(from Petroleum Engineering 412 Course Notes -- 1997)

Petroleum Engineering 324
Well Performance

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Pseudosteady-State Flow in a Radial System

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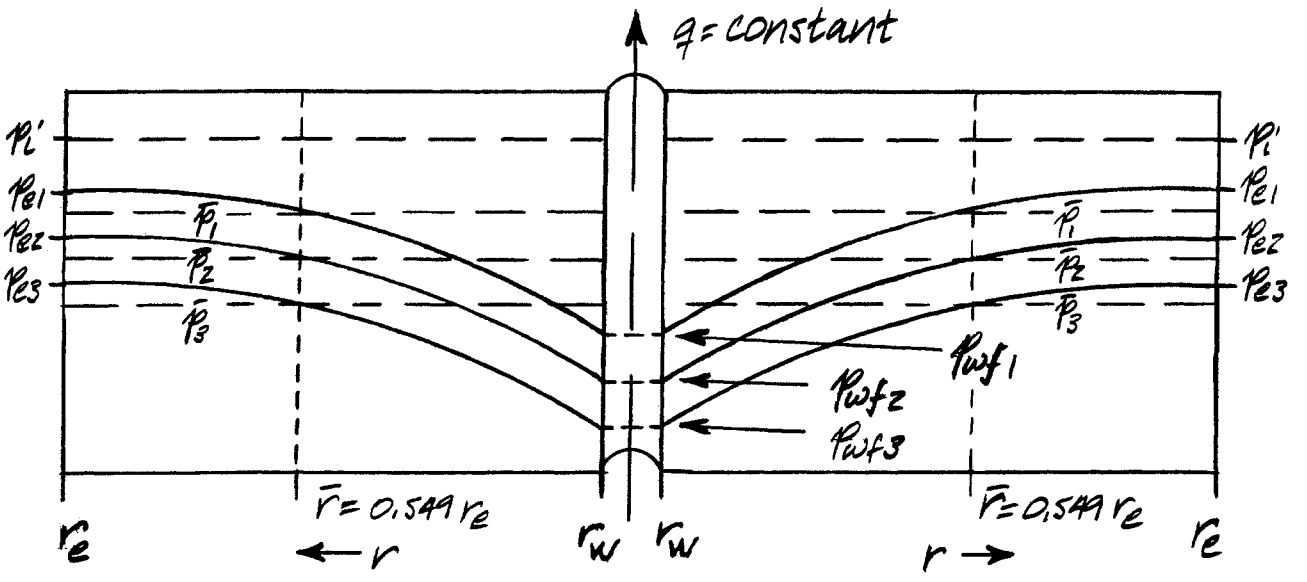
Physical Considerations

The physical concept of pseudosteady-state is defined as the condition where the pressure at all points in the reservoir changes at the same rate. Mathematically, this condition is given by:

$$\frac{d}{dt}\left[p(r,t)\right]\Big|_r = \text{constant} \qquad (1)$$

Physically, this condition is illustrated by



where,

p_{wfi} = wellbore pressure at time, t_i

$\bar{p}_i$ = average reservoir pressure at time, t_i

p_{ei} = external boundary pressure at time, t_i

Objectives: (for pseudosteady-state flow conditions)

1. Derive a pressure change relation (i.e., $d\bar{p}/dt$) using the material balance relation.
2. Derive a relation between the average reservoir pressure, $\bar{p}$, and the wellbore flowing pressure, p_{wf}.
3. Derive a pressure, radius, time (i.e., $p(r,t)$) solution of the radial flow diffusivity equation.

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Material Balance Considerations

Recalling the material balance relation for a slightly compressible liquid, we have

$$\bar{p} = p_i - \frac{B}{NB_i c_t} N_p \qquad (2)$$

or, noting that $NB_i = V_p$ we obtain

$$\bar{p} = p_i - \frac{B}{V_p c_t} N_p \qquad (3)$$

For a cylindrical reservoir, we have

$$V_p = \phi h \pi (r_e^2 - r_w^2) \qquad (4)$$

Substituting Eq. 4 into Eq. 3 gives us

$$\bar{p} = p_i - \frac{B}{\phi h \pi (r_e^2 - r_w^2) c_t} N_p \qquad (5)$$

Recalling the definition of the cumulative production, N_p, we have

$$N_p = \int_0^t q(\tau)\, d\tau \qquad (6)$$

Therefore,

$$\frac{dN_p}{dt} = q \qquad (7)$$

Taking the derivative of Eq. 5 with respect to time

$$\frac{d\bar{p}}{dt} = -\frac{B}{\phi h \pi (r_e^2 - r_w^2) c_t} q \qquad (8)$$

Note: all derivations are in "Darcy" Units unless otherwise noted.

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Pseudosteady-State Flow Solutions for the Radial Flow Diffusivity Equation

The governing partial differential equation for flow in porous media is called the "diffusivity" equation. The diffusivity equation for a "slightly compressible liquid" is given (without derivation) as

$$\frac{1}{r}\frac{\partial}{\partial r}\left[r\frac{\partial p}{\partial r}\right] = \frac{\phi \mu c_t}{k}\frac{\partial p}{\partial t} \qquad (9)$$

The significant assumptions made in Eq. 9 are:

- slightly compressible liquid (constant compressibility)
- constant fluid viscosity
- single-phase liquid flow
- gravity and capillary pressure are neglected
- constant permeability
- horizontal radial flow (no vertical flow)

If we assume that the flowrate, q, is constant then $d\bar{p}/dt$ is also constant—hence $\partial p/\partial t$ is constant as well. Assuming q is constant, then

$$\frac{d\bar{p}}{dt} = \frac{\partial p}{\partial t} = -\frac{B}{\phi h \pi (r_e^2 - r_w^2) c_t} q = \text{constant} \qquad (10)$$

Substituting Eq. 10 into Eq. 9 (we note that partial derivatives are now expressed as ordinary derivatives), this gives

$$\frac{1}{r}\frac{d}{dr}\left[r\frac{dp}{dr}\right] = \left[\frac{\phi \mu c_t}{k}\right]\left[-\frac{B}{\phi h \pi (r_e^2 - r_w^2) c_t} q\right]$$

or, reducing

$$\frac{1}{r}\frac{d}{dr}\left[r\frac{dp}{dr}\right] = -\frac{qB\mu}{\pi k h (r_e^2 - r_w^2)} \qquad (11)$$

Defining

$$c = \frac{qB\mu}{\pi kh(r_e^2 - r_w^2)} \tag{12}$$

Substituting Eq. 12 into Eq. 11 we have

$$\frac{1}{r}\frac{d}{dr}\left[r\frac{dp}{dr}\right] = -c \tag{13}$$

Separating

$$d\left[r\frac{dp}{dr}\right] = -c\,r\,dr$$

Integrating (indefinite integration)

$$\int d\left[r\frac{dp}{dr}\right] = -c\int r\,dr$$

Completing

$$r\frac{dp}{dr} = -\frac{c}{2}r^2 + c_1 \tag{14}$$

Multiplying through Eq. 14 by $1/r$ gives us

$$\frac{dp}{dr} = -\frac{c}{2}r + c_1\frac{1}{r} \tag{15}$$

For pseudosteady-state we assume a closed reservoir, that is

$$\left[\frac{dp}{dr}\right]_{r_e} = 0$$

or

$$\left[\frac{dp}{dr}\right]_{r_e} = 0 = -\frac{c}{2}r_e + c_1\frac{1}{r_e}$$

Solving for c_1 gives

$$c_1 = \frac{c}{2}r_e^2 \tag{16}$$

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Substituting Eq. 16 into Eq. 15 gives

$$\frac{dp}{dr} = \frac{c}{2}\left[\frac{r_e^2}{r} - r\right] \qquad (17)$$

Multiplying through Eq. 17 by dr gives us

$$dp = \frac{c}{2}\left[\frac{r_e^2}{r} - r\right] dr$$

Integrating across the reservoir, we have

$$\int_{p_{wf}}^{p_r} dp = \frac{c}{2}\int_{r_w}^{r}\left[\frac{r_e^2}{r} - r\right] dr \qquad (18)$$

Completing the integration

$$p_r - p_{wf} = \frac{c}{2}\left[r_e^2 \ln(r)\Big|_{r_w}^{r} - \frac{r^2}{2}\Big|_{r_w}^{r}\right]$$

or

$$p_r - p_{wf} = \frac{c}{2}\left[r_e^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{2}(r^2 - r_w^2)\right] \qquad (19)$$

Recalling Eq. 12

$$c = \frac{qB\mu}{\pi kh(r_e^2 - r_w^2)} \qquad (20)$$

Substituting Eq. 20 into Eq. 19, we obtain

$$p_r - p_{wf} = \frac{qB\mu}{2\pi kh}\,\frac{1}{(r_e^2 - r_w^2)}\left[r_e^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{2}(r^2 - r_w^2)\right]$$

Expanding through with the $1/(r_e^2 - r_w^2)$ term gives

$$p_r - p_{wf} = \frac{qB\mu}{2\pi kh}\left[\frac{r_e^2}{(r_e^2 - r_w^2)} \ln\left[\frac{r}{r_w}\right] - \frac{1}{2}\frac{(r^2 - r_w^2)}{(r_e^2 - r_w^2)}\right] \qquad (21)$$

Eq. 21 is our final result (in "Darcy" units).

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Development of a $\bar{p}$-p_{wf} Relation - Pseudostate-State Flow

In this section we develop the relationship between the average reservoir pressure, $\bar{p}$, and the wellbore flowing pressure, p_{wf}. The definition of the average reservoir pressure is given as

$$\bar{p}_r \equiv \frac{\int_{r_w}^{r} p_r \, dV}{\int_{r_w}^{r} dV} \qquad (22)$$

and for a cylindrical reservoir, we have

$$V = \phi h \pi (r^2 - r_w^2) \qquad (23)$$

$$dV = \phi h \pi (2r) \, dr \qquad (24)$$

Substituting Eq. 24 into Eq. 22 gives

$$\bar{p}_r = \frac{\phi h 2\pi}{\phi h \pi (r^2 - r_w^2)} \int_{r_w}^{r} p_r \, r \, dr$$

which reduces to

$$\bar{p}_r = \frac{2}{(r^2 - r_w^2)} \int_{r_w}^{r} p_r \, r \, dr \qquad (25)$$

Solving Eq. 21 for p_r gives us

$$p_r = p_{wf} + \frac{q B \mu}{2\pi k h} \left[\frac{r_e^2}{(r_e^2 - r_w^2)} \ln\left[\frac{r}{r_w}\right] - \frac{1}{2} \frac{(r^2 - r_w^2)}{(r_e^2 - r_w^2)} \right] \qquad (26)$$

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Substituting Eq. 26 into Eq. 25 gives

$$\bar{p}_r = \frac{2}{(r^2-r_w^2)}\int_{r_w}^{r}\left[p_{wf} + \frac{qB\mu}{2\pi kh}\left[\frac{r_e^2}{(r_e^2-r_w^2)}\ln\left[\frac{r}{r_w}\right] - \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)}\right]\right] r\,dr \qquad (27)$$

Separating

$$\bar{p}_r = \frac{2}{(r^2-r_w^2)}\, p_{wf}\int_{r_w}^{r} r\,dr$$

$$+ \frac{2}{(r^2-r_w^2)}\,\frac{qB\mu}{2\pi kh}\,\frac{r_e^2}{(r_e^2-r_w^2)}\int_{r_w}^{r} r\ln\left[\frac{r}{r_w}\right] dr$$

$$- \frac{2}{(r^2-r_w^2)}\,\frac{qB\mu}{2\pi kh}\,\frac{1}{2(r_e^2-r_w^2)}\int_{r_w}^{r} r^3\,dr$$

$$+ \frac{2}{(r^2-r_w^2)}\,\frac{qB\mu}{2\pi kh}\,\frac{r_w^2}{2(r_e^2-r_w^2)}\int_{r_w}^{r} r\,dr \qquad (28)$$

Isolating terms and evaluating each integral, we have

$$\int_{r_w}^{r} r\,dr = \frac{1}{2}(r^2-r_w^2) \qquad (29)$$

$$\int_{r_w}^{r} r^3\,dr = \frac{1}{4}(r^4-r_w^4) \qquad (30)$$

$$\int_{r_w}^{r} r\ln\left[\frac{r}{r_w}\right] dr = ?$$

Obviously, the integral of the logarithm term will require a little work to resolve. We could simply look up the appropriate result in a suitable text, – but deriving the required result will be enlightning.

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Starting with the fundamental form of the logarithm integral, we have

$$\int x \ln(x/c)\,dx$$

... integration by parts $\quad \int u\,dv = uv - \int v\,du$

$$u = \ln(x/c) \qquad dv = x\,dx$$

$$du = \frac{1}{x}\,dx \qquad v = \frac{1}{2}x^2$$

Then

$$\int x \ln(x/c)\,dx = \frac{1}{2}x^2 \ln(x/c) - \frac{1}{2}\int x\,dx$$

Reducing

$$\int x \ln(x/c)\,dx = \frac{1}{2}x^2 \ln(x/c) - \frac{1}{4}x^2$$

Therefore

$$\int_{r_w}^{r} r \ln\left[\frac{r}{r_w}\right] dr = \left[\frac{1}{2}r^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{4}r^2\right]_{r_w}^{r}$$

$$= \frac{1}{2}r^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{4}r^2 - \left[\underbrace{\frac{1}{2}r_w^2 \ln\left[\frac{r_w}{r_w}\right]}_{0} - \frac{1}{4}r_w^2\right]$$

or finally, we have

$$\int_{r_w}^{r} r \ln\left[\frac{r}{r_w}\right] dr = \frac{1}{2}r^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{4}(r^2 - r_w^2) \qquad (31)$$

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Substituting Eqs. 29-31 into Eq. 28 gives

$$\bar{p}_r = \frac{2}{(r^2-r_w^2)} p_{wf} \frac{1}{2}(r^2-r_w^2)$$

$$+ \frac{2}{(r^2-r_w^2)} \frac{qB\mu}{2\pi kh} \frac{r_e^2}{(r_e^2-r_w^2)} \left[\frac{1}{2} r^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{4}(r^2-r_w^2) \right]$$

$$- \frac{2}{(r^2-r_w^2)} \frac{qB\mu}{2\pi kh} \frac{1}{2(r_e^2-r_w^2)} \frac{1}{4}(r^4-r_w^4)$$

$$+ \frac{2}{(r^2-r_w^2)} \frac{qB\mu}{2\pi kh} \frac{r_w^2}{2(r_e^2-r_w^2)} \frac{1}{2}(r^2-r_w^2)$$

Reducing

$$\bar{p}_r = p_{wf} + \frac{qB\mu}{2\pi kh} \frac{2}{(r^2-r_w^2)} \frac{r_e^2}{(r_e^2-r_w^2)} \left[\frac{1}{2} r^2 \ln\left[\frac{r}{r_w}\right] - \frac{1}{4}(r^2-r_w^2) \right]$$

$$- \frac{qB\mu}{2\pi kh} \frac{2}{(r^2-r_w^2)} \frac{1}{2(r_e^2-r_w^2)} \frac{1}{4}(r^2-r_w^2)(r^2+r_w^2)$$

$$+ \frac{qB\mu}{2\pi kh} \frac{2}{(r^2-r_w^2)} \frac{1}{2(r_e^2-r_w^2)} \frac{r_w^2}{2}(r^2-r_w^2)$$

Collecting

$$\bar{p}_r = p_{wf} + \frac{qB\mu}{2\pi kh} \frac{r_e^2}{(r_e^2-r_w^2)} \left[\frac{r^2}{(r^2-r_w^2)} \ln\left[\frac{r}{r_w}\right] - \frac{1}{2} \right]$$

$$- \frac{qB\mu}{2\pi kh} \frac{(r^2+r_w^2)}{4(r_e^2-r_w^2)} + \frac{qB\mu}{2\pi kh} \frac{r_w^2}{2(r_e^2-r_w^2)}$$

or "finally"

$$\bar{p}_r = p_{wf}$$

$$+ \frac{qB\mu}{2\pi kh} \left[\frac{r_e^2}{(r_e^2-r_w^2)} \left[\frac{r^2}{(r^2-r_w^2)} \ln\left[\frac{r}{r_w}\right] - \frac{1}{2} \right] - \frac{(r^2+r_w^2)}{4(r_e^2-r_w^2)} + \frac{r_w^2}{2(r_e^2-r_w^2)} \right] \quad (32)$$

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Eq. 32 (which is given in "Darcy" units) is our fundamental linking relation between the wellbore and average reservoir pressures during pseudosteady-state flow. However, $\bar{p}_r$ (the average reservoir pressure at a given radius, r) is of little use - except as a rigorous "linking" relation for pressures in the reservoir.

In contrast, if we consider $\bar{p}_{re}$ (i.e., p_r at $r = r_e$) we obtain the average reservoir pressure based on the entire reservoir volume. Such a result can be directly coupled with the material balance equation to develop a time-pressure relation for pseudosteady-state flow.

Evaluating Eq. 32 at $r = r_e$ we have

$$\bar{p} \equiv \bar{p}_{re} = p_{wf} + \frac{qB\mu}{2\pi kh}\left[\frac{r_e^2}{(r_e^2 - r_w^2)}\left[\frac{r_e^2}{(r_e^2 - r_w^2)}\ln\left[\frac{r_e}{r_w}\right] - \frac{1}{2}\right] - \frac{(r_e^2 + r_w^2)}{4(r_e^2 - r_w^2)} + \frac{r_w^2}{2(r_e^2 - r_w^2)}\right] \quad (33)$$

Assuming that $r_e >> r_w$, then

$$\frac{r_e^2}{(r_e^2 - r_w^2)} \approx 1 \; ; \quad \frac{(r_e^2 + r_w^2)}{(r_e^2 - r_w^2)} \approx 1 \; ; \quad \frac{r_w^2}{(r_e^2 - r_w^2)} \approx 0$$

Substituting these expressions into Eq. 33, we obtain

$$\bar{p} = p_{wf} + \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{1}{2} - \frac{1}{4}\right]$$

or

$$\bar{p} = p_{wf} + \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] \quad (34)$$

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Summarizing our results so far (using generalized units systems)

Pressure at any Radius:

$$p_r = p_{wf} + \frac{qB\mu}{c_r kh}\left[\frac{r_e^2}{(r_e^2-r_w^2)}\ln\left[\frac{r}{r_w}\right] - \frac{(r^2-r_w^2)}{2(r_e^2-r_w^2)}\right] \qquad (35)$$

Average Reservoir Pressure at any Radius:

$$\bar{p}_r = p_{wf} + \frac{qB\mu}{c_r kh}\left[\frac{r_e^2}{(r_e^2-r_w^2)}\left[\frac{r^2}{(r^2-r_w^2)}\ln\left[\frac{r}{r_w}\right]-\frac{1}{2}\right] - \frac{(r^2+r_w^2)}{4(r_e^2-r_w^2)} + \frac{r_w^2}{2(r_e^2-r_w^2)}\right] \qquad (36)$$

Average Reservoir Pressure at r_e (Volumetric Average Pressure):

$$\bar{p} = p_{wf} + \frac{qB\mu}{c_r kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] \qquad (37)$$

For a general reservoir geometry, Eq. 38 becomes

$$\bar{p} = p_{wf} + \frac{qB\mu}{c_r kh}\left[\frac{1}{2}\ln\left[\frac{4}{e^{\gamma}}\frac{A}{r_w^2}\frac{1}{C_A}\right]\right] \qquad (38)$$

where

$\gamma = 0.577216...$ Euler's Constant

C_A = Dietz "shape factor" (e.g. $C_A = 31.62$ for circular reservoir)

Table of Units Conversion Factors

Factor	Darcy Units	Field Units	SI Units
c_r	2π	$2\pi \times 1.127\times10^{-3}$ or 7.081×10^{-3}	$2\pi \times 8.527\times10^{-5}$ or 5.358×10^{-4}

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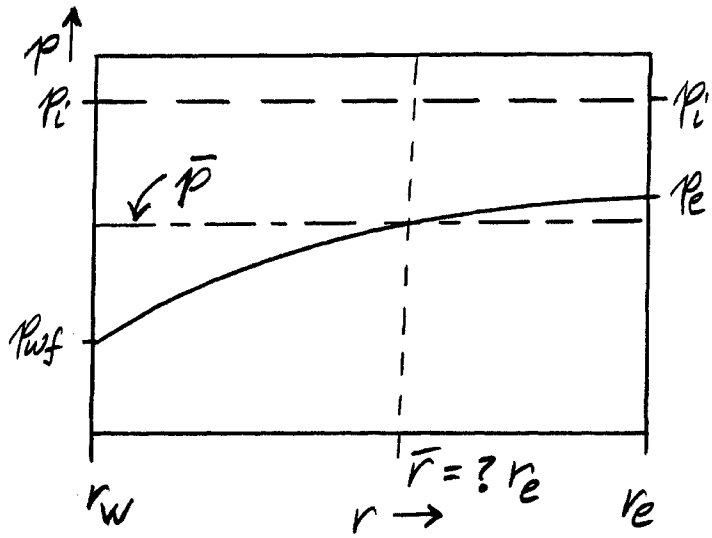
$\bar{p}$ - $\bar{r}$ Concept

An interesting (and possibly useful) result is the concept of $\bar{r}$, which would be the location of the average reservoir pressure, $\bar{p}$. This development is only rigorously valid for a vertical well centered in a bounded circular reservoir. A graphical illustration of this concept is shown below.



Mathematically, $\bar{r}$ is defined by equating the p_r relation (Eq. 35) with the average reservoir pressure identity, $\bar{p}$, (Eq. 37). Equating Eqns. 35 and 37 gives

$$\frac{r_e^2}{(r_e^2 - r_w^2)} \ln\left[\frac{\bar{r}}{r_w}\right] - \frac{(\bar{r}^2 - r_w^2)}{2(r_e^2 - r_w^2)} = \ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}$$

We now assume $\frac{r_e^2}{(r_e^2 - r_w^2)} \approx 1$ and $\frac{r_w^2}{(r_e^2 - r_w^2)} \approx 0$ gives

$$0 = \ln\left[\frac{\bar{r}}{r_w}\right] - \ln\left[\frac{r_e}{r_w}\right] - \frac{\bar{r}^2}{2(r_e^2 - r_w^2)} + \frac{3}{4}$$

Assuming that $r_e \gg r_w$ (i.e. $(r_e^2 - r_w^2) \approx r_e^2$) and rearranging, we have

$$\ln\left[\frac{\bar{r}}{r_e}\right] - \frac{\bar{r}^2}{2r_e^2} + \frac{3}{4} = 0 \qquad (39)$$

Defining a dimensionless radius, $\bar{r}_D$, we obtain

$$\bar{r}_D = \frac{\bar{r}}{r_e} \qquad (40)$$

Substituting Eq. 40 into Eq. 39 gives us

$$\ln(\bar{r}_D) - \frac{1}{2}\bar{r}_D^2 + \frac{3}{4} = 0 \qquad (41)$$

Solving Eq. 41 for $\bar{r}_D$, we obtain

$$\bar{r}_D = 0.54928\ldots \quad (\text{or } \bar{r} = 0.54928\ldots r_e) \qquad (42)$$

<u>Development of a $p(r,t)$ Relation for Pseudosteady-State Flow</u>

Our last objective is to develop a $p(r,t)$ relation for pseudosteady-state flow in a bounded circular reservoir. Recalling the material balance relation (Eq. 5) we have

$$\bar{p} = p_i - \frac{B}{\phi h \pi (r_e^2 - r_w^2) c_t} N_p \qquad (5)$$

For a constant flowrate, q, we have

$$N_p = \int_0^t q(\tau)\,d\tau = qt \qquad (43)$$

Substituting Eq. 43 into Eq. 5,

$$\bar{p} = p_i - \frac{qB}{\phi h \pi (r_e^2 - r_w^2) c_t} t \quad (\text{Darcy units}) \qquad (44)$$

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Recalling the average reservoir pressure identity for a well centered in a bounded circular reservoir, we have

$$\bar{p} = p_{wf} + \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] \quad \text{(Darcy Units)} \qquad (45)$$

Substituting Eq. 45 into Eq. 44 gives

$$p_{wf} + \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] = p_i - \frac{qB}{\phi h\pi(r_e^2 - r_w^2)c_t}\,t$$

Rearranging

$$p_i - p_{wf} = \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] + \frac{qB}{\phi h\pi(r_e^2 - r_w^2)c_t}\,t \qquad (46)$$

or

$$p_i - p_{wf} = \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] + \frac{qB}{V_p c_t}\,t \qquad (47)$$

Recalling the wellbore-reservoir pressure relation (Eq. 26), we have (upon slight rearranging)

$$p_r - p_{wf} = \frac{qB\mu}{2\pi kh}\left[\frac{r_e^2}{(r_e^2 - r_w^2)}\ln\left[\frac{r}{r_w}\right] - \frac{1}{2}\frac{(r^2 - r_w^2)}{(r_e^2 - r_w^2)}\right] \qquad (48)$$

Subtracting Eq. 48 from Eq. 47 and solving for p_r gives us

$$p_r = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4} - \frac{r_e^2}{(r_e^2 - r_w^2)}\ln\left[\frac{r}{r_w}\right] + \frac{1}{2}\frac{(r^2 - r_w^2)}{(r_e^2 - r_w^2)}\right] - \frac{qB}{V_p c_t}\,t \qquad (49)$$

Assuming that $r_e \gg r_w$ (i.e., $(r_e^2 - r_w^2) \approx r_e^2$) gives us

$$p_r = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r}\right] + \frac{1}{2}\frac{(r^2 - r_w^2)}{(r_e^2 - r_w^2)} - \frac{3}{4}\right] - \frac{qB}{V_p c_t}\,t \qquad (50)$$

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Summarizing, we have the following relations in <u>Darcy units</u>

$$p_r = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4} - \frac{r_e^2}{(r_e^2-r_w^2)}\ln\left[\frac{r}{r_w}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)}\right] - \frac{qB}{V_p c_t}t \quad (49)$$

and

$$p_r = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)} - \frac{3}{4}\right] - \frac{qB}{V_p c_t}t \quad (50)$$

In <u>Field Units</u> we have

$$p_r = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4} - \frac{r_e^2}{(r_e^2-r_w^2)}\ln\left[\frac{r}{r_w}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)}\right]$$

$$- 5.615\frac{qB}{V_p c_t}t \quad (t \text{ in days}, V_p \text{ in ft}^3) \quad (51)$$

and

$$p_r = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r}\right] + \frac{1}{2}\frac{(r^2-r_w^2)}{(r_e^2-r_w^2)} - \frac{3}{4}\right] - 5.615\frac{qB}{V_p c_t}t \quad (52)$$

(t in days, V_p in ft^3)

For t in hours we use $5.615/24 = 0.23395$.

Finally for conditions at the well, we have

<u>Darcy Units</u>: $$p_{wf} = p_i - \frac{qB\mu}{2\pi kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] - \frac{qB}{V_p c_t}t \quad (53)$$

<u>Field Units</u>: (t, days; V_p, ft^3) $$p_{wf} = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] - 5.615\frac{qB}{V_p c_t}t \quad (54)$$

(t, hours; V_p, ft^3) $$p_{wf} = p_i - 141.2\frac{qB\mu}{kh}\left[\ln\left[\frac{r_e}{r_w}\right] - \frac{3}{4}\right] - 0.23395\frac{qB}{V_p c_t}t \quad (55)$$

Recall that the pore volume, V_p, is given by

$$V_p = \phi h\pi(r_e^2 - r_w^2) = \phi hA \quad (4)$$